Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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Table of Contents

Summary 3

Introduction and Project overview 4

Methodology 5

Data Cleaning and Preparation 5

Data Visualization and Findings 6

Predictive modelling 6

Insights 7

Conclusion 7

References 8

Summary

This report presents a complete data science workflow built for a publishing analytics team. The goal was to collect, clean, explore, and analyse book product listings from Books to Scrape (books.toscrape.com). Using Python and BeautifulSoup, all 1,000 books across 50 web pages were scraped. Data cleaning removed formatting errors and handled missing values. Exploratory data analysis showed that prices are spread evenly between £10 and £60, with a mean price of £35.07. A Linear Regression model was built to predict prices, which confirmed that ratings and title lengths do not influence book prices in this dataset.

Introduction and Project overview

Modern online stores use data analysis to set prices, manage inventory, and understand customer choices. This project simulates the real-world role of a Junior Data Scientist tasked with analysing an online book catalogue. The target site is Books to Scrape, a standard website used for web scraping practice.

- 1000 Total books scraped
- 50 pages processes
- £35,07 is the average price
- £35,98 is the median price

Key Objectives:

- Automatically download the book information using BeautifulSoup web scraping.
- Clean raw text data into numbers, handle missing data, and create useful metrics.
- Create charts to understand book prices and ratings
- Build a linear regression model to test if book features can predict retail prices.

Methodology

The dataset was collected directly from Books to Scrape. The website holds 1,000 books spread across 50 pages (20 books per page). A Python script using requests and BeautifulSoup was written to loop through all 50 pages automatically.

The scraper successfully processed page 1/50 through page 50/50 without dropping connections, saving a total of 1,000 complete records. This met and exceeded the assignment requirement of at least 200 books.

Data Cleaning and Preparation

Raw scraped data usually contains extra symbols and missing values that cannot be used directly in machine learning models. The following cleaning steps were applied to prepare the dataset:

Field Name	Original Format	Cleaned Format	Cleaning Method
Price	String (“£51,77”)	Numeric (51,77)	Removed currency symbol “£” and converted to float.
Rating	Text(“Three”)	Integer (3)	Mapped words (“One” to “Five”) to numbers (1 to 5).
Stock quantity	Text (“In stock (22 available)”)	Integer (22)	Extracted numbers using regular expressions
Description	2 Missing Values	Clean Text	Replaced missing text with blank space to keep rows

The title length and description word count were created to see if the text length relates to the price.

Data Visualization and Findings

Exploratory data analysis was performed using Seaborn and Matplotlib to inspect price patterns, stock levels, and star ratings across the book catalogue.

Price Distribution Findings:

- Book prices range evenly between £10.00 and £50.00+.
- The overall mean price is £35.07 and the median price is £35.98.
- The distribution histogram shows a flat, uniform spread, which indicates that prices on this practice website were generated randomly.
- Star ratings (1 star to 5 stars) are also evenly split across all books (~200 books per rating level).

Predictive modelling

A Multiple Linear Regression model was trained to predict book prices using star ratings, stock quantities, title lengths, and description word counts. The dataset was split into 80% for training (800 books) and 20% for testing (200 books).

Model Metric	Score	Explanation
R-Squared	0,00	0% of the price changes
MAE	£12,45	On average, predictions are off by £12,45
RMSE	£14,38	RMSE measures the average size of prediction errors.

Model Conclusion: Star ratings and text length have no statistical relationship with book prices in this dataset. This happens because the sandbox website assigned prices at random during creation.

Insights

Based on the analysis of the 1,000-book catalogue, here are practical recommendations for online bookstores, publishers, and data analysts:

- For Online Retailers: Ensure every book has a complete product description. Missing metadata makes books harder to find in search engines. Dynamic pricing should rely on real competitor prices rather than internal text lengths.
- For Publishers: Since star ratings do not affect book prices, publishers can price high-quality special edition books higher without fear that lower rating counts will force price cuts.
- For Data Analytics Teams: To improve future price prediction models, add extra data such as book binding format (hardcover vs. paperback), author fame, page numbers, and real sales numbers.

Conclusion

This project successfully demonstrated an end-to-end data science pipeline for web scraping, data cleaning, visualization, and modeling. All 1,000 book records were safely scraped and cleaned into a complete analytical CSV file. While the linear regression model confirmed that surface details do not predict synthetic book prices, the automated scraping pipeline and data cleaning methods provide a reliable framework for real-world market analysis.

References

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